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UNIT B WORKBOOK



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KEY



Chapter 5: Energy Transformation

Define the following:

Energy

ability to do work (p. 82)

Potential Energy

stored energy that can be converted into other forms (p. 82)

Kinetic Energy

energy due to motion (p. 82)

Solar Energy

energy from the Sun (p. 84)

Thermal Energy

heat (p. 84)

Wind Energy

kinetic energy captured by windmills (p. 84)

Geothermal Energy

heat produced within Earth (from geothermal activity) that come close to the surface; can be used for heating or to produce electricity (p. 84)

Input Energy

energy that enters the system (p. 86)

Converter

system in which energy is changed from one form into another (p. 86)

Output Energy

useful energy that leaves the system (p. 86)

Chemical Energy

energy in chemicals;

for example, potential energy from food converts to chemical energy during digestion and chemical energy in fireworks ingredients is transformed to light, sound, and heat energy when the fireworks are set off (p. 87)

1. Law of Conservation of Energy

energy cannot be created or destroyed, only converted from one form to another; input energy is equal to output energy (p. 92)

2. List 3 examples of kinetic energy. (pg. 82)

Swinging a golf club

Golf ball flying through the air

Walking

Sliding down a hill

3. List 3 examples of potential energy. (pg. 82)

Food stores energy

Sitting on top of a hill

Batteries store energy

A spring compressed in a mouse trap

4. List the energies you depend on when you are: (pgs. 82, 84 & 85)

a. Walking to school

Chemical potential energy stored in body

Kinetic energy of the motion of legs

b. Driving to the store

Chemical potential energy stored in gasoline

Kinetic energy of the motion of car engine and motion of the car

c. Turning on the computer

Chemical potential energy stored in body

Kinetic energy of motion of fingers flicking a switch

Kinetic energy of the motion of electrons in the computer

d. Listening to the radio

Kinetic energy of the movement of the speaker and the motion of molecules in the air

5. List two ways the Sun's energy is converted for use. (pgs. 84-85)

- **by solar panels for electricity**
- **by chlorophyll in plants to manufacture sugar**
- **by deer by eating plants for energy**
- **by windmills to make electricity**

6. Is thermal energy a form of kinetic energy? Explain. (pg. 84)

Yes.

Thermal energy involves the motion of molecules that move faster and faster.

7. What energy conversions take place from the eating of an apple to the energy used to throw a ball? (pgs. 84-85)

**solar energy → chemical potential energy → conversion to a form cells can use
→ muscular activity**

- **Photosynthesis involves the conversion of solar energy into glucose, a form of chemical potential energy.**
- **Cellular respiration harvests the energy from glucose by breaking it down chemically into a form that cells can use.**
- **In the human body, this chemical potential energy is used to drive chemical reactions and do mechanical work such throwing a ball.**

8. Briefly explain how plants and animals use energy from the Sun. (pg. 84)

Plants:

- **absorb the energy of the Sun through the chlorophyll in their leaves**
- **use energy to produce glucose**
- **convert the chemical energy in glucose into other forms of energy that the plant uses for its life processes; i.e., cellular respiration**

Animals:

- **eat plants and break down their glucose**
- **convert the chemical energy in glucose into other forms of energy that the animal uses for its life processes; i.e., cellular respiration**

9. List 8 forms of energy and give 2 examples of each. (pg. 85)

Solar (Light)

Potential (Stored Energy – Potential to do work/cause movement)

- Gravitational (due to height – potential to fall)
- Chemical (energy stored in matter)
- Elastic/spring (stretched or compressed objects)

Kinetic – energy of motion

Thermal-Electric Generating Stations

(produces electricity with heat boiling water to create pressure to turn turbines)

- Burning fossil fuels (Coal, Oil, Natural Gas)
- Nuclear

Hydro-Electric (produces electricity with falling/moving water turning turbines)

Tidal (produces electricity with falling/moving water turning turbines)

Wind (produces electricity with wind turning turbines)

Geothermal (produces electricity with steam pressure turning turbines)

Electrical (moving electrons in wires)

Sound (waves of compressed air)

Thermal (moving atoms/molecules)

10. What 3 components are necessary for an energy conversion? (pg 86)

input energy, converter, output energy

11. Identify the energy conversions that take place in the following scenarios.

a. When the inside of a car heats up when the car is left in the Sun.

input (solar energy enters car through glass) →

**converter (solar energy changes frequency as it goes through glass —
it can get in but not out) →**

output (the car heats up)

b. When you paddle a canoe

input – chemical potential energy in body

converter - muscles

output – kinetic energy of paddles and canoe moving

c. A model rocket taking off

input – chemical potential energy in fuel

converter – rocket engine

output – rocket lifts off

d. Drill press cutting a piece of wood

input – electrical energy

converter – drill press motor

output – drill spins and cuts wood

12. List 3 examples of energy conversion in which heat is given off. (pgs. 86 to 91)
- **cellular respiration**
 - **burning of wood on a campfire**
 - **burning of fuel from a butane lighter**
 - **burning of natural gas in a furnace**
13. Give two examples of waste thermal energy that might cause a problem? (pg. 91)
- **on extremely hot days, when athletes may get overheated.**
 - **Thermal energy produced by many machines often interferes with operation.**
14. List 3 kinds of energy that come from a blow dryer. (pg. 92)
- **thermal**
 - **mechanical/kinetic**
 - **sound**
 - **light**
15. Most of the energy produced by a car engine is not used to make the car move.
Where does the rest of the energy go? (pg. 92)
- The majority of this energy is released into the environment as thermal energy**
16. Use an example to explain the Law of Conservation of Energy. (pg. 92)
- Any answer showing all energy input is accounted for as output energy is correct.**
- 1. When a car stops, all the kinetic energy the car had is transferred.**
- **Some of the energy goes to the brakes. As the brake pads press against the drums to make the vehicle stop, friction creates thermal energy.**
 - **As the car stops, the wheels stop moving. This creates friction between the tires and pavement. This friction produces thermal energy.**
 - **Air resistance is also involved. As the car stops, the air in front of it holds it back, assisting in the stop.**
- The air friction between the car and the air also uses energy.**
- 2. When a table saw or blender is running, all the electric energy that enters the machine is balanced by energy that leaves the machine.**
- **Some energy goes to turn the machine.**
 - **Some energy is converted to thermal energy caused by friction, both in the motor and in the blades.**
 - **Some energy goes to sound.**
 - **The balance of the energy does the work of cutting or mixing.**

17. Explain why input energy is greater than useful output energy. (pg. 94)

Some energy is always lost in the form of thermal energy or due to friction

18. Sketch two machines that have not been discussed but you are familiar with.

On the sketch, identify where there is waste output energy. (pg. 94)

Accept a variety of sketches. Look for the following criteria:

- **waste output energy**

- This might appear as thermal energy where fuel is burned, or where friction occurs between moving parts.

- **amounts of waste energy**

- If students include electronic machines such as a computer or calculator, they might make some mention of the small amount of waste energy.

There is a warm-up time and such machines do get warm, so waste energy should be shown coming from the top.

Chapter 6: Generating Electric Energy

Define the following:

Generator

device that makes electricity (p. 100)

Turbine

sets of blades put at an angle so they can collect energy from fastmoving water, steam, or wind (p. 102)

Hydroelectric

kinetic energy of moving water converted into electric energy (page 102)

Thermoelectric

chemical energy stored in a fossil fuel and converted into thermal energy, then to kinetic energy, then to electric energy (p. 102)

Thermonuclear

thermal energy released when atoms are split that is used to produce electricity (p. 102)

Solar Cells

items that convert light energy (usually energy from the Sun) into electricity (p. 107)

Newton

N; unit of measure for force; force applied over distance (p. 108)

Power

rate at which energy is transferred or work is done (p. 108)

Watt

W; international unit representing one joule of energy per second (p. 108)

Watt Hour (W*H)

W•h; one watt of power per hour; see kW•h (p. 109)

Kilowatt Hours (kW*h)

kW•h; 1000 watts per hour (p. 110)

Efficiency

measure of how well energy is converted to accomplish the intended task (p. 112)

1. Draw and label the main parts of a generator. (pg. 101)

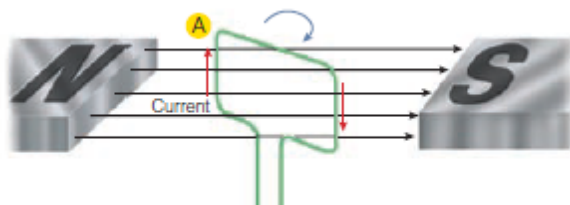


Figure 6.4 A generator consists of a magnet and a coil of wire. Spinning the coil of wire through the magnetic field is one way to generate electricity.

2. Describe how a simple generator produces electric energy. (pgs. 100 & 101)

The two main parts of a generator are a magnet and a coil of wire.

Moving a wire coil through a magnetic field produces electricity.

This can be achieved by moving the coil around the magnet or by moving the magnet through the coil.

3. List two things you can do to increase the amount of electric energy produced by a generator. (pgs. 100 & 101)

- **use stronger magnets**
- **use more wire**
- **use thinner wire**
- **use a more conductive type of wire**

4. List 3 ways energy is generated in Canada. (pg. 102)

In Canada, electricity is generated mainly by:

- **converting kinetic energy of moving water (in a hydro-electric plant)**
- **converting chemical energy stored in fossil fuels (in a thermo-electric plant)**
- **converting energy released when atoms are split (in a thermonuclear plant)**

5. Describe how coal is used to produce electricity. (pg. 104)

Pulverized coal is burned in a furnace.

Heat (thermal energy) from the burning coal converts water in a boiler to steam, which turns the blades of a turbine.

The turbine turns a generator, which converts the mechanical energy into electric energy.

6. List two advantages of generating electric energy using a hydroelectric dam. (pg. 105)

Advantages:

- There are no harmful emissions.
- Hydro power is efficient. The dams last for a long time.
- After the dam is completed, there is little environmental damage.
- Generators run with little maintenance.

Disadvantages:

- Hydro power dams cause major environmental disruption or damage.
- Dams hold back sediments, thus increasing erosion of the downstream riverbed.
- Farmland that depends on flooding for fertility shows less productivity.
- Moving people to make way for hydro reservoirs causes social disruption or a lot of person grief, as people must move from places they may have inhabited for generations.
- Moving people to make way for hydro reservoirs is a particular problem in areas (such as northern sections of Canada and rural sections of India) where people live traditionally because the move may destroy their method of making a living from the land.

7. What are some possible ecological damages that may result because of a hydroelectric dam? (pg. 105)

- land and water ecosystems upstream and downstream of the dam are disrupted or destroyed;
- reservoirs may flood human settlements, forests, and agricultural land;

8. How much energy is lost as electricity moves from generating stations to your home? How is it lost? (pg. 106)

About 10 percent is lost.

As it passes through the lines, it is converted to thermal energy and wasted.

9. List 3 steps that take place as nuclear energy is converted to electric energy. (pg. 106)

Answers will vary, but should include three steps from the following sequence.

- Uranium fuel rods are inserted into the reactor core, and nuclear fission begins.
- Fission reactions produce thermal energy, which is carried away by a coolant.
- Coolant is taken to a tank where the thermal energy converts water into steam.
- Steam is directed through the blades of a turbine. The blades turn, spinning a shaft.
- Spinning turbine turns a generator, which converts mechanical energy into electric energy.
- Transmission lines take electric energy to customers.

KEY

10. What does a thermo-electric plant have in common with a thermonuclear power plant?
(pg. 106)

In both types of power plants, thermal energy is converted to electric energy using steam and turbines.

11. What is power? (pg. 108)

Power is the rate at which energy is transferred.

12. What are the units used to measure the power of electric appliances?(pgs. 109 & 110)

watts

1 watt = 1 joule per second (1 W = 1 J/s)

13. Calculate the following examples using the formula
(REMEMBER time should be in seconds).

$$\text{Power} = \frac{\text{energy}}{\text{time}}$$

- a. 60 joules of energy are used in 2 hours

Energy = 60 J

Time = 2 h = 7200 s

$$\text{power} = \frac{\text{energy}}{\text{time}} = \frac{60 \text{ J}}{7200 \text{ s}} = 0.0083 \frac{\text{J}}{\text{s}} = 0.0083 \text{ W}$$

- b. 120 joules in half a minute

Energy = 120 J

Time = 0.5 min = 30 s

$$\text{power} = \frac{\text{energy}}{\text{time}} = \frac{120 \text{ J}}{30 \text{ s}} = 4 \frac{\text{J}}{\text{s}} = 4 \text{ W}$$

- c. 200 joules in 50 seconds

Energy = 200 J

Time = 50 s

$$\text{power} = \frac{\text{energy}}{\text{time}} = \frac{200 \text{ J}}{50 \text{ s}} = 4 \frac{\text{J}}{\text{s}} = 4 \text{ W}$$

14. What are the units for the energy we buy from the power company. (pg. 110)

energy = power x time = kW x h
kilowatt hours (kW•h)

15. Using the formula (energy = power x time) calculate the following examples.

a. A 100W light bulb is left on for 4 hours.

$$\begin{array}{ll} \text{Power} = 100 \text{ W} & \text{energy} = \text{power} \times \text{time} \\ \text{Time} = 4 \text{ h} & = 100 \text{ W} \times 4 \text{ h} = 400 \text{ W}\cdot\text{h} \\ & = 0.4 \text{ kW}\cdot\text{h} \end{array}$$

b. A 240 Watt fan is left on for 2 hours.

$$\begin{array}{ll} \text{Power} = 240 \text{ W} & \text{energy} = \text{power} \times \text{time} \\ \text{Time} = 2 \text{ h} & = 240 \text{ W} \times 2 \text{ h} = 480 \text{ W}\cdot\text{h} \\ & = 0.48 \text{ kW}\cdot\text{h} \end{array}$$

c. A 540 W stove is left on for 3 hours.

$$\begin{array}{ll} \text{Power} = 540 \text{ W} & \text{energy} = \text{power} \times \text{time} \\ \text{Time} = 3 \text{ h} & = 540 \text{ W} \times 3 \text{ h} = 1620 \text{ W}\cdot\text{h} \\ & = 1.62 \text{ kW}\cdot\text{h} \end{array}$$

16. What is the difference between a kWh and a Wh? (pgs. 109 & 110)

One kilowatt hour is equivalent to 1000 watt hours.

17. Why would the electric company use kilowatt hours instead of watt hours? (pg. 110)

The watt hour is too small a unit of energy.

The larger unit (kW•h) is more convenient because numbers can be expressed more simply (e.g. 35 500 W•h vs. 35.5 kW•h).

18. How do we calculate efficiency? (pg. 112)

You can determine the efficiency of an electric device by dividing useful energy output by the total energy input, and then multiplying by 100.

Efficiency is always expressed as a percent.

$$\text{Percent efficiency of an electric device} = \frac{\text{useful energy output}}{\text{total electric energy input}} \times 100$$

19. Why does an incandescent light bulb get so much hotter than a fluorescent light bulb? (pg. 111)

The filament in an incandescent bulb has a high resistance, which makes the filament glow white-hot.

This produces light energy, but also produces a large amount of thermal energy (heat).

20. Use an example to define "efficiency". (pgs. 111 to 113)

Answers might be worded in the following fashion:

**Only 10 percent of the total electric energy input to an incandescent light bulb is converted to light energy;
the rest is converted to thermal energy (heat).**

Since the bulb's intended purpose is for use as a light, the heat is considered waste energy.

Therefore the efficiency of the incandescent light bulb is 10 percent, which is extremely inefficient.

21. State 3 ways to conserve energy at home. (pg. 113)

- **turn off lights and television when not in use;**
- **take short showers to save hot water;**
- **keep thermostat at a low setting — wear extra layers for warmth instead;**
- **install energy-efficient appliances;**
- **use compact fluorescent light bulbs instead of incandescent light bulbs;**
- **caulk around window panes to conserve heat during the winter;**
- **use draft barriers along bottoms of doors;**
- **shut doors to rooms that are not being used;**
- **plant trees to shade the house in summer and protect it from cold winds in winter;**
- **close drapes in summer to keep out heat; open them in winter during sunny days.**

Chapter 7: Energy for Life

Define the following:

Chloroplasts

special parts of green plants that capture solar energy (p. 118)

Chlorophyll

green enzyme in plant cells that enables photosynthesis (p. 118)

Photosynthesis

process by which plants use chlorophyll and sunlight to convert carbon dioxide and water into glucose and oxygen (p. 118)

Cellular Respiration

process used by all plant and animal cells to obtain energy from food (p. 119)

Homeostasis

state of equilibrium or balance; regulation of the body's internal environment (p. 123)

Metabolism

rate at which a human body uses energy (p. 123)

Carbohydrate

main source of energy for the human body; usually in the form of glucose (p. 125)

Glycogen

starch-like carbohydrate in which the body stores glucose for future use (p. 125)

Fat

source of energy and energy storage in animals; see also *saturated fats* and *unsaturated fats* (p. 126)

Saturated Fats

animal fats and some oils; associated with an increased risk of heart disease (p. 126)

Unsaturated Fats

most vegetable oils; associated with a lowered risk of heart disease (p. 126)

Protein

building block and source of energy for the body (p. 127)

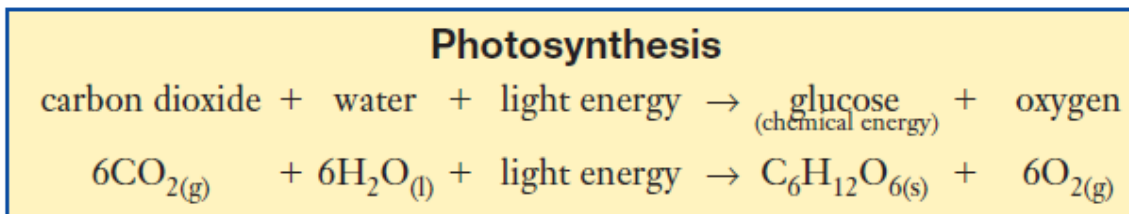
Metabolic needs

body's needs for energy (p. 130)

1. Where in plants does photosynthesis take place? (pg. 118)

Chloroplasts

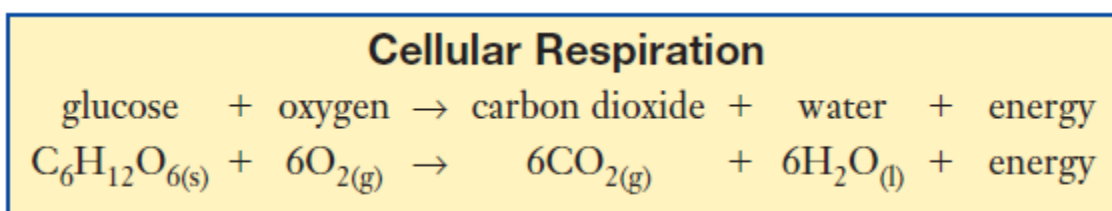
2. Write the formula for photosynthesis. (pg. 118)



3. Where does cellular respiration take place? (pg. 119)

Mitochondria

4. Write the formula for cellular respiration. (pg. 119)



5. What is the main source of energy of all the cells in our body? (pg. 119)

glucose

6. What would happen if we had no mitochondria in our bodies? (pg. 119)

We would not have the ability to convert energy from glucose into useable energy for our bodies

7. Why is cellular respiration important? (pg. 119)

Cellular respiration is a process used by all plant and animal cells to obtain energy from food.

During this process, energy stored in glucose combines with oxygen to produce carbon dioxide, water, and energy

8. What does lactic acid do to your muscles? (pg. 119)

When lactic acid is produced, muscles tire quickly, and become stiff and painful.

KEY

9. How do animals capture energy from the sun? (pg. 122)

Our bodies cannot capture energy directly from the Sun.

Energy is transferred to our bodies when we eat green plants or when we eat the meat of animals that have eaten green plants.

10. As you go higher on the pyramid of energy, is there more or less energy available? Why is this so? (pg. 122)

Less and less energy is available as you get higher on the pyramid.

This happens because energy transformations are never 100 percent efficient and much of the energy is transformed into thermal energy at each level.

Thermal energy cannot be used by organisms at the next level of the food chain.

- **From level to level, 10 percent of the initial available energy is passed on to be used by the next level.**
- **90 percent of the energy is released as waste thermal energy, used for growth and reproduction, or stays in the organism.**

11. Why are there less organisms as you move up a food chain? (pg. 122)

It takes many organisms at a lower level to support a small number of organisms at a higher level.

This is because of the amount of waste energy at each level.

12. How much energy is transferred to the next level in a food chain? Pg 122

10%

13. Draw a food chain that is at least 4 animals long. Label the Producer, Primary consumer, Secondary consumer and the tertiary consumer . (pg. 122)

Grain → Mice → Snake → Owl
(producer) (primary (secondary (tertiary
consumer) consumer) consumer)

14. What is the internal balance in the body called? (pg. 123)

Homeostasis

15. List 2 things that your body will do to cool itself down? (pg. 123)

- **Blood vessels are brought close to the skin's surface.**
Heat can then transfer out quickly.
- **Perspiration is another way the body cools.**
Moisture is moved outside the skin.
Evaporation of this moisture cools the body.

16. List 2 things your body does to warm itself up? (pg. 123)

- **Shivering helps to warm your body. Vibrating muscles burn more energy.**
The chemical potential energy in glucose is converted to the kinetic energy used in shivering, and thermal energy is released.
- **To protect itself from getting too cold, the human body can direct less blood to limbs and more to the core.**
To do this, blood vessels near the surface constrict. Less blood can get close to the outside, so less thermal energy is lost to the environment.

17. List 2 adaptations that the Inuit have to keep them warm. (Internet)

- **Eat a high fat marine mammal diet**
- **Have a high metabolism that produces body heat**

18. List 2 adaptations that desert people have that allows them to survive. (Internet)

The desert-adapted person can sweat freely but must deal with the water loss involved. This adaptation minimizes both water needs and water loss.

Skin pigmentation is moderate since extreme pigmentation is good protection from the sun but allows absorption of heat, which must be lost by sweating.

19. How does homeostasis and metabolism work together to keep an organism alive? (pg. 124)

- **Homeostasis describes an organism's ability to maintain a balance or steady-state.**
- **Metabolism describes the rate at which a body uses energy.**
- **There are mechanisms that maintain a human body's temperature around 37°C.**
As metabolism increases, mechanisms are in place to remove the excess thermal energy produced and cool the body.
- **If body temperature drops below the normal level, metabolic activities increase, which releases more thermal energy to warm the body.**

20. What is the main source of energy for the human body? (pg. 125)

Carbohydrates

21. What role do carbohydrates play in cellular respiration? (pg.125)

Carbohydrates are the main source of energy for cellular respiration.

22. What percent of our daily calories should be carbohydrates? (pg. 125)

Health experts recommend 55% to 65% of daily calories from carbohydrates

23. What is the difference between glycogen and glucose? How do they work together? (pg. 125)

Through cellular respiration, glucose is broken down to provide energy that powers the body's metabolism.

If more glucose is taken in than is immediately needed, the extra is converted to glycogen, a starch-like substance.

Glycogen is stored in the liver and muscles.

When we need energy, glycogen is converted back into glucose.

24. List 4 carbohydrates. (pg. 125)

rice, pasta, potatoes, pizza, fruit, bread, baked beans, yogurt.

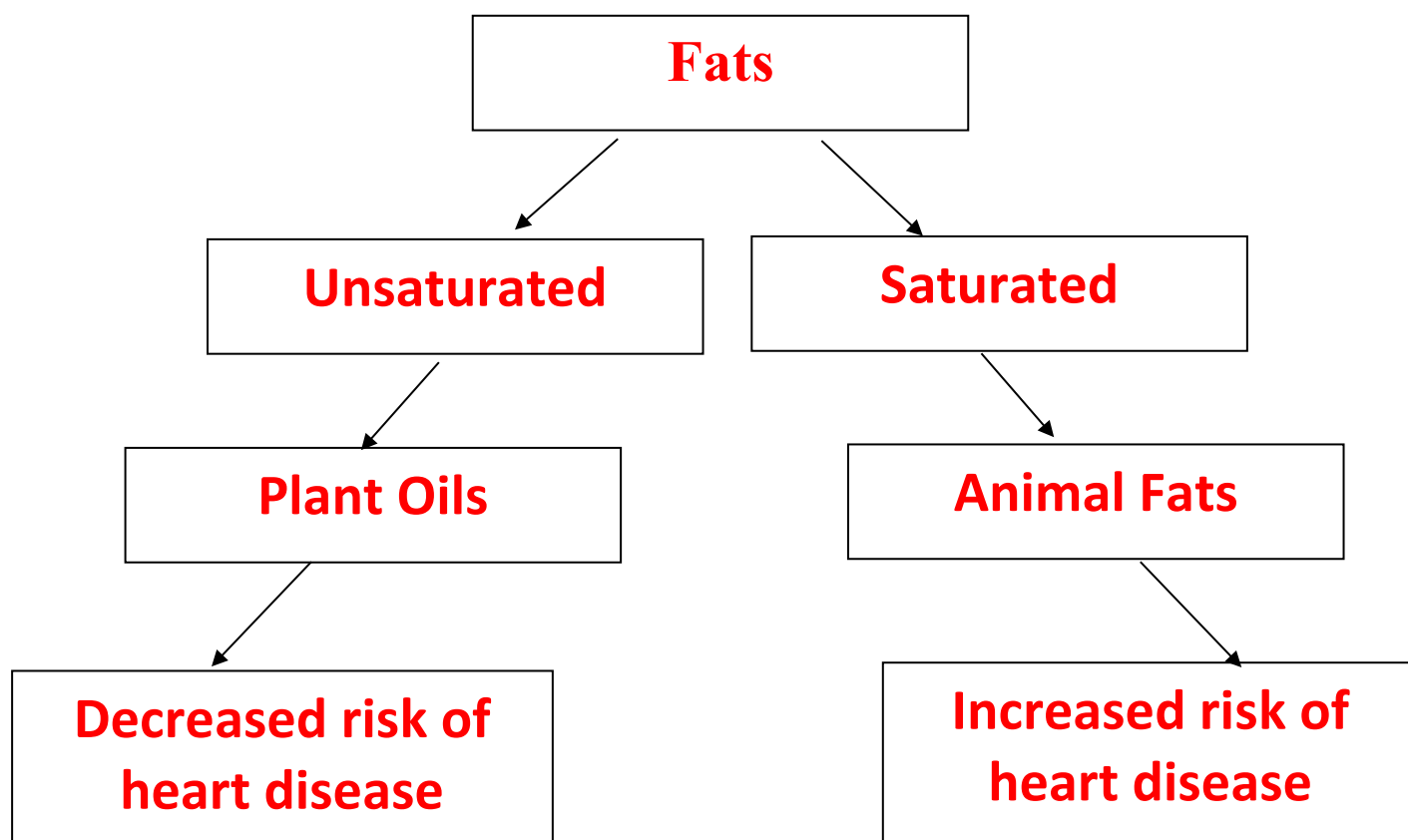
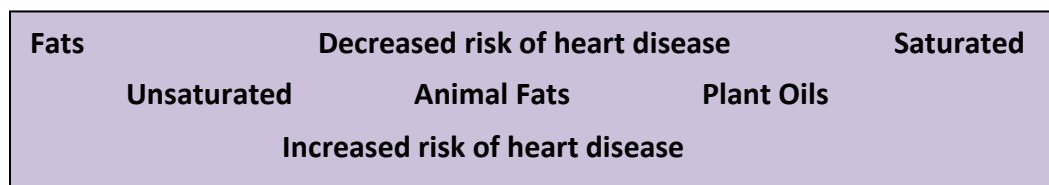
25. What do we need fat for? (pg.126)

We need fat for insulation, to conserve energy, and as a source of energy.

26. List 4 sources of fat. (pg. 126)

animal fats, vegetable oils, dairy, nuts

27. Complete the following flow chart using the following words.



28. What is the difference between protein and carbohydrates? (pg. 127)

One gram of protein contains the same energy as one gram of carbohydrate. The difference is that it is harder for the body to break it down to release the energy, so it takes longer.

That is why the body will use protein as an energy source only when there is little carbohydrate or fat left.

29. Why is protein important for our body? (pg. 127)

Proteins help build and maintain body tissues. They are used for growth.

KEY

30. List 4 sources of protein. (pg. 127)
Meat, cheese, milk, eggs, beans
31. Why do children need more protein than adults? (pg. 127)
Proteins help build and maintain body tissues. They are used for growth.
32. In what situations would your body use protein for energy? (pg. 127)
The body uses proteins for energy when fats and carbohydrates are not available, i.e., in conditions of starvation.
33. In your own words define metabolism? (pgs. 130 & 131)
Metabolism is the sum of all of the chemical reactions taking place in the body.
34. Describe some of the differences in nutrient and energy requirements between a baby and a senior citizen? (pg. 130)
 - **Babies have greater nutritional and energy requirements because of their high rate of growth and fast metabolism.**
 - **Senior citizens have lower nutritional and energy requirements since they have slower metabolisms and their growth has stopped.**
35. List 3 occupations in which you would expect workers to have high metabolisms? (pg. 131)
Careers that involve high activity levels, such as professional athlete, construction worker, firefighter, dancer, etc
36. For each of the following situations, predict whether the person's metabolic rate will go up, down, or stay the same.
- a) A construction worker resigns and takes on a rural mail route. **Go down**
 - b) A mother stops breast feeding. **Go down**
 - c) A high school gymnast begins training for the provincial finals **Go up**
 - d) A weightlifter takes time off from training **Go down**

Chapter 8: Fossil Fuels

Define the following:

Fossil Fuels

formed from the remains of swamp vegetation and/or animals that lived about 300 million years ago; include coal, crude oil (petroleum), and natural gas (p. 136)

Coal

see *fossil fuels* (p. 137)

Natural gas

see *fossil fuels* (p. 137)

Crude Oil

petroleum; see *fossil fuels* (p. 137)

Petroleum

crude oil; see *fossil fuels* (p. 137)

Reservoir rock

porous rock containing tiny droplets of oil forced there by changes in Earth's crust (p. 140)

Seismic survey

method used by geologists to determine the location of potential oil and natural gas reservoirs by sending shock waves through the layers of Earth's crust that reflect off rock;

the printout from this survey shows the underground rock formations;

see *reservoir rock* (p. 140)

Pump jack

working component in an oil well; lifts oil located deep underground hundreds of meters to the surface (p. 141)

Lift pump

working component in an oil well located deep underground; lifts oil hundreds of meters to the surface (p. 141)

Distillation

process by which a substance (such as crude oil) is separated into its component parts by boiling it and then condensing the resulting vapors (p. 141)

Combustion

chemical reaction in which a substance combines with oxygen to produce heat and light; burning (pp. 28, 144)

Hydrocarbons

group of fossil fuels so named because they are made up of carbon and hydrogen (p. 144)

Non-renewable

useable only once; will not form again in nature in the foreseeable future (p. 147)

Acid deposition

acidic materials (liquids or solids that contain sulfuric or nitric acid) that are deposited on Earth (pp. 64, 148)

Global climate change

Formerly called global warming; during this process, some places on Earth may get much warmer while others may get cooler; see *climate change* (p. 149)

1. List 3 fossil fuels. (pg. 137)

Coal, oil natural gas

2. Sketch and LABEL four steps in the formation of coal. (pg. 136)

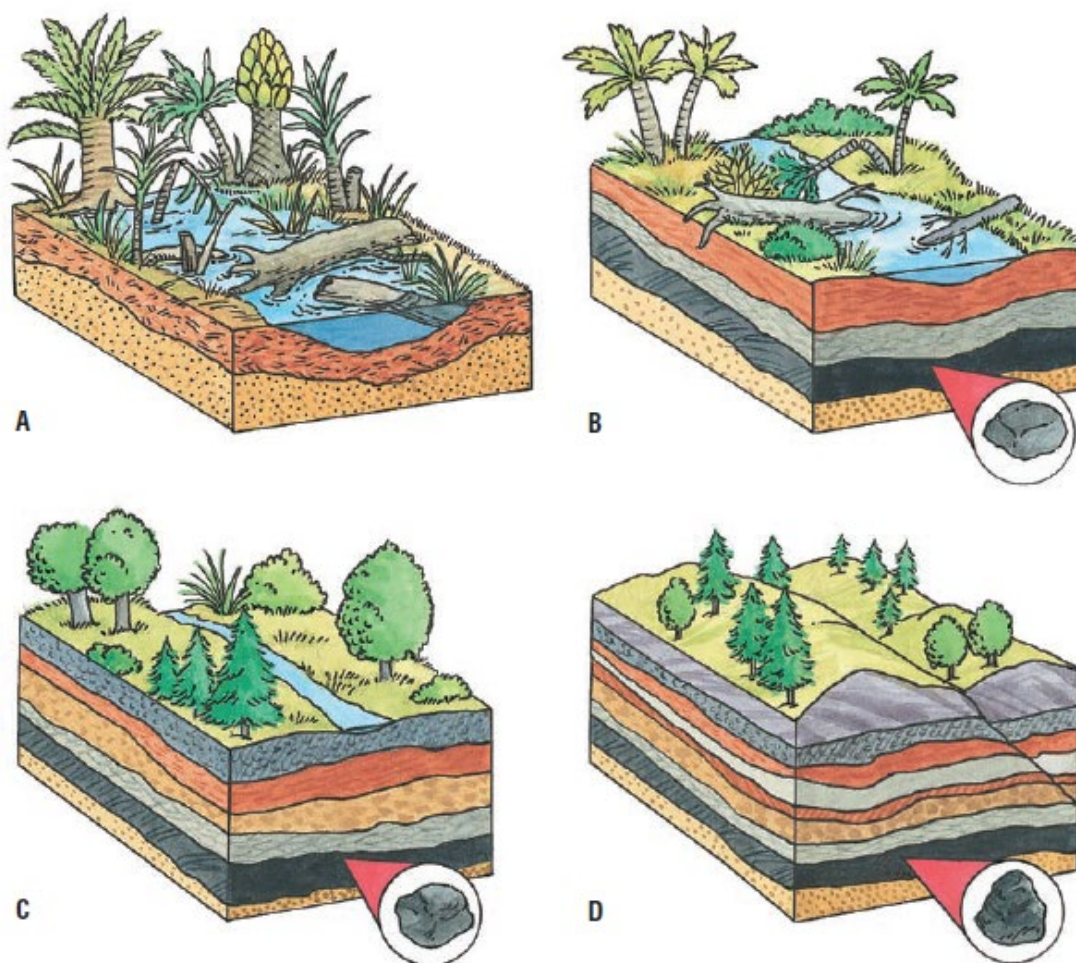


Figure 8.2 Fossil fuels take a long time to develop. These visuals show how different types of coal developed. Crude oil was formed in a similar way.

- A** Coal was formed from the remains of ancient swamp vegetation that lived about 300 million years ago.
- B** The plants died and fell into the swamp. Their organic material was covered by layers of sediment. There was no oxygen in these layers, so the plant material could not decompose. Hydrogen and carbon from the organic material were locked together underground. Eventually, they formed a solid layer of lignite, which is a soft coal.
- C** More layers of sediment piled on top of the lignite and compressed it even further. Because the layers were so deep and there was so much pressure, temperatures increased. The heat and pressure compressed the lignite into a harder coal, called bituminous coal.
- D** In some areas, layers of bituminous coal were compressed and heated even further. This produced an even harder coal called anthracite.

3. Describe two ways in which coal is mined. (pg. 138)

- **surface mining/open-pit mines**
- **underground tunnels**

4. List 3 uses for fossil fuels.(pg. 137)

Coal	Oil	Natural Gas
• generate electricity • heat houses and factories • manufacture iron • manufacture steel	• generate electricity • heat houses and factories • power vehicles	• heat houses and factories • generate electricity • power vehicles

5. Explain how the various components of crude oil can be separated. (pg. 143)

- **Crude oil is vaporized using steam heat.**
- **The vapour is cooled at different levels in the fractional or distillation tower.**
- **Liquids are collected at different parts of the tower according to the boiling point of each.**

6. Sketch how oil, water and natural gas arrange themselves under the Earth's surface. (pg. 140)

Look for the following order (top to bottom): natural gas, water, oil.

Students should explain that natural gas is a vapour so it seeks the highest possible position.

Water floats on oil, so will be second, and oil is the heaviest, so will be deepest.

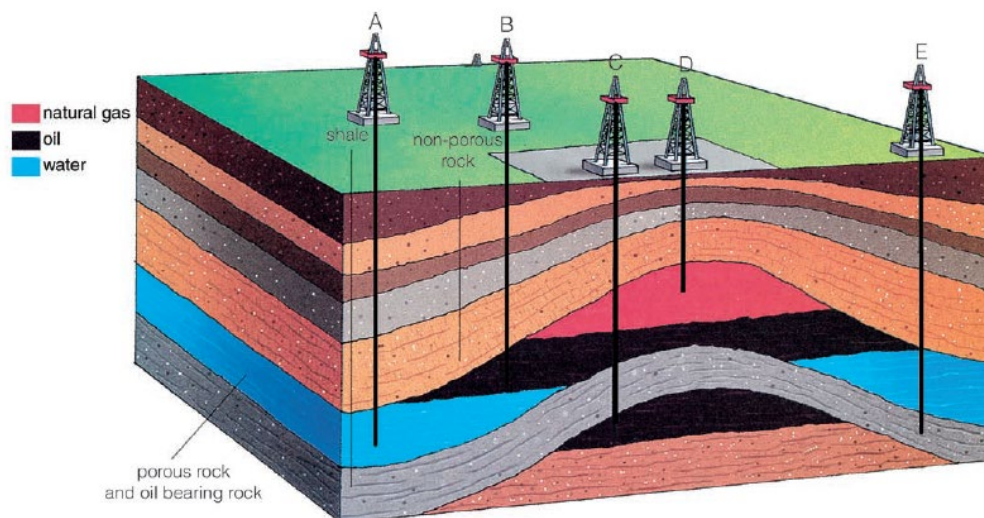


Figure 8.7 As oil developed, it mixed with water and sand. This mixture, along with bubbles of natural gas, seeped through porous layers of rock. Eventually, the mixture reached a layer of rock it could not pass through and was trapped.

The water, oil, and natural gas gathered under the non-porous layer of rock. Because natural gas is less dense than oil, it rose to the top.

The well at D pumps natural gas. What do the wells at A, B, C, and E get?

7. Explain how shockwaves are used to find oil.(pg. 140)

Shock waves or seismic waves are caused by explosions.

They travel through the ground and reflect off rocks in the subsurface.

Special electronic equipment records the energy reflected back and uses it to produce a picture of the underground rock formations.

8. How are pumps used to bring oil to the surface?(pg. 141)

Pumps are used to bring oil to the surface when the pressure of the underground oil is not strong enough to bring it to the surface naturally.

Here is how a hand pump works:

- 1. When the piston is raised (upstroke), a movable plate at the top of the pump is raised.
The valve at the bottom of the pump opens and the valve on the top plate closes. This allows water to enter the lower chamber.**
- 2. When the piston is pushed down, the plate in the pump is also pushed back down.
The bottom valve closes and the liquid in the lower chamber is forced through the open valve into the upper chamber.**
- 3. On the next upstroke of the plate, the liquid that had passed through the top valve spills out the spout.**

Figure 1 Piston is moving up

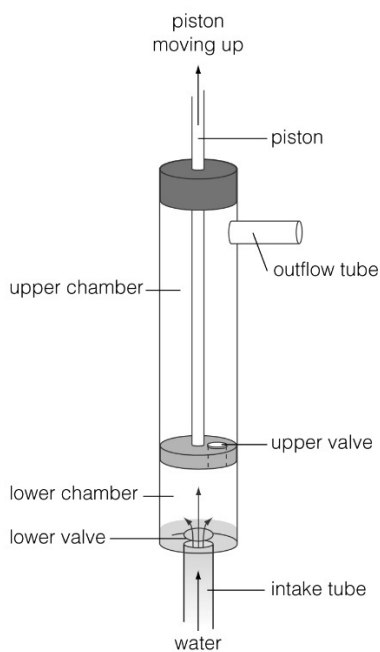
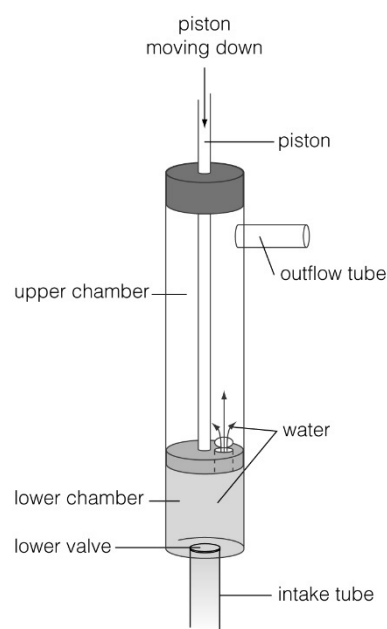


Figure 2 Piston is moving down



KEY

9. List 3 technologies that use combustion to convert the energy stored in fossil fuels into other forms of energy. (pg. 138-143)

Steam engines, furnaces, lamps, stoves, automobile engines, generate electricity

10. Copy the word and chemical equations for the combustion of propane. $C_3H_8(g)$.
Label the input energy, converter, and output energy. (pg. 144)

propane gas + oxygen gas $\rightarrow$ carbon dioxide gas + water vapour



11. Identify the two gases produced by both combustion and cellular respiration. (pg. 146)

Carbon dioxide and water vapour

12. How else is cellular respiration similar to fossil fuel combustion? (pg. 146)

Comparison Point	Cellular Respiration	Combustion
Is oxygen a reactant?	yes	yes
What are the products?	$CO_{2(g)} + HOH_{(l)}$	$CO_{2(g)} + HOH_{(l)}$
Is thermal energy released?	yes	yes
Is there a chemical change?	yes	yes

- **Oxygen is used in each case.**
- **Thermal energy is released in both reactions.**
- **There is a chemical change in each case.**

13. Why are fossil fuels said to be non-renewable? (pg. 147)

Once used, they are gone.

They cannot be re-used and will not form again within the foreseeable future. Just like fossil fuels from the past, plants and animals that die today will take hundreds of millions of years to become fossil fuels — if they ever do.

14. List 3 risks associated with our use of fossil fuels.(pg. 147 & 149)

- **burning fossil fuels causes smog**
- **burning fossil fuels causes or speeds up global warming**
- **burning fossil fuels causes acid deposition**
- **transporting crude oil is accompanied by risks of spills of poisonous substances**

KEY

15. Use a labeled diagram or flowchart to explain how an increase in greenhouse gas might make Earth warmer. (pg. 62 and 149)

The diagram or chart should show:

- **burning of fossil fuels;**
- **amount of carbon dioxide in atmosphere increasing; and**
- **solar energy entering but being hindered as it tries to leave Earth.**

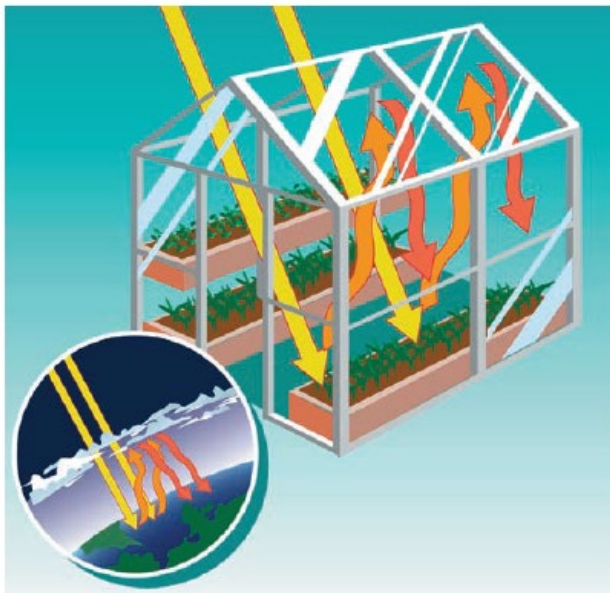


Figure 4.2 A greenhouse prevents heat from escaping into the atmosphere. Greenhouse gases work the same way to trap heat in Earth's atmosphere.

16. Why is global climate change a concern? (pg. 149)

- **Earth's climate zones may change.**
- **species cannot adapt to fast changes in climate**
- **As ice caps and glaciers melt, low-lying areas will flood, sea levels rise**